

**INTERSECONDARY SCHOOLS EXAMINATION SERIES
ISESE**



FORM TWO PRE-NECTA SERIES No.01

031

PHYSICS

TIME 2:30 HRS

Year 2026

INSTRUCTIONS

1. This paper consists of sections A and B with a total of **ten (10)** questions.
2. Answer **all** questions in the spaces provided.
3. Section A and C carry **fifteen (15)** marks each and section B carries **seventy (70)** marks.
4. All writing must be in **black** or **blue** ink **except** diagrams which must be in pencil.
5. Communication devices and any unauthorized materials are **not** allowed in the assessment room.
6. Write your **name** at every page.

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**FORM TWO AND FORM FOUR PRE NECTA SERIES
ISESE TIME TABLE FOR GROUP II
FOR REGISTRATION CALL US 0624 254 757**

ALL SUBJECTS	DATE
SERIES - 01	03 rd August 2026
SERIES - 02	10 th August 2026
SERIES - 03	17 th August 2026
SERIES - 04	24 th August 2026
SERIES - 05	31 st August 2026

NB# SERIES No.06 – No.10 (SEPTEMBER – OCTOBER)

SECTION A (15 Marks)

Answer **all** questions in this section.

1. For each of the items (i) - (x), choose the correct alternative from among the given options and write its letter in the answering space provided.

(i) A laboratory technician wants to store a highly volatile and flammable chemical liquid safely. Which of the following standard actions best reflects an understanding of safety rules regarding the introductory physics laboratory environment?

- A. Storing the chemical near a window where sunlight illuminates the label clearly.
- B. Keeping the container tightly sealed on a shelf away from any open burner flames.
- C. Leaving the bottle unstoppered inside the fume cupboard to prevent pressure build-up.
- D. Placing the chemical on the floor right next to the main exit door for quick disposal.

(ii) While experimenting with a standard measuring instrument, a student notices that the pointer does not align with the zero mark when no load or quantity is being measured. This type of error is categorized as:

- A. Parallax error due to wrong positioning of the observer's eye.
- B. Environmental error caused by sudden changes in room temperature.
- C. Systematic error inherent to the calibration or condition of the instrument.
- D. Random error that can be completely eliminated by taking many averages.

(iii) When an astronaut travels from the Earth to the Moon, their physical characteristics change relative to the new environment. Which statement correctly describes what happens to their mass and weight?

- A. Both mass and weight decrease due to the weaker gravitational pulling force.
- B. The mass remains completely unchanged, while the weight decreases.
- C. The weight remains exactly the same, but the mass experiences a noticeable decrease.
- D. Both mass and weight remain completely constant regardless of location.

(iv) A local artisan mixes two different liquid substances together to create a synthetic lubricant. In determining the overall density of the resulting uniform mixture, which fundamental factor must be considered?

- A. The total volume of the mixture divided by the difference in the individual boiling points.
- B. Only the density of the heavier liquid substance regardless of proportions.
- C. The total combined mass of the substances divided by their total combined volume.

D. The product of the individual densities divided by the atmospheric pressure.

(v) A heavy cargo ship made of dense steel floats perfectly on the ocean, whereas a small solid steel bolt quickly drops to the bottom of a water bucket. What explains this difference in sinking and floating behavior?

A. The ship's unique structural design encloses a vast volume of air, making its average density less than that of water.

B. Sinking depends entirely on the absolute weight of an object rather than its volume.

C. Ocean water lacks any upthrust force compared to the water found in small domestic buckets.

D. The steel bolt experiences a much larger buoyant force than the massive cargo ship.

(vi) Engineers designing a modern suspension bridge must select materials that can withstand severe stretching forces without breaking or permanently deforming. Which mechanical property of matter is most critical for this structural requirement?

A. High brittleness to ensure the material snaps instantly when overloaded.

B. High elasticity and tensile strength to recover original dimensions after stress.

C. Low ductibility so that the material cannot change its shape under any condition.

D. High plasticity to allow the bridge anchors to permanently deform every day.

(vii) A person wearing high-heeled pointed shoes steps on a soft muddy lawn and sinks deeply, whereas another person of the exact same weight wearing flat-soled shoes walks across the same lawn without sinking. This scenario demonstrates that:

A. Pressure is directly proportional to the total surface area of contact.

B. The total downward force applied to the ground depends on the type of footwear.

C. Pressure increases significantly when a given force is distributed over a smaller surface area.

D. Flat shoes generate a much greater localized downthrust than pointed shoes.

(viii) A worker pushes a heavy wooden crate horizontally across a rough factory floor at a constant speed, covering a distance of ten meters. In the context of physics, work is said to be done because:

A. The worker spent a long time thinking about the best way to move the crate.

B. A force was applied, and the crate experienced a displacement in the direction of that force.

C. The gravitational potential energy of the crate increased dramatically during motion.

D. The crate remained completely stationary despite the large effort applied.

(ix) A dry plastic comb is run through dry hair several times on a cold afternoon. When the comb is brought near tiny scraps of paper on a table, the paper bits are instantly attracted to it. This phenomenon is primarily caused by:

- A. The creation of a temporary magnetic field within the plastic material.
- B. The accumulation of stationary electric charges on the comb's surface through friction.
- C. Continuous current electricity flowing from the hand into the paper pieces.
- D. The sudden alignment of magnetic domains inside the pieces of paper.

(x) When a beam of white solar light passes through a triangular glass medium under precise conditions, it splits neatly into a beautiful band of distinct colors ranging from red to violet. This phenomenon is known as:

- A. Regular reflection from highly polished transparent glass surfaces.
- B. Total internal reflection where light cannot escape into the air.
- C. Dispersion, which occurs because different colors of light travel at different speeds in glass.
- D. Optical magnification by an instrument designed to enlarge distant objects.

ANSWER

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match each description of an optical component or function in **LIST A** with its corresponding term or feature from **LIST B** by writing the letter of the correct response.

LIST A	LIST B
(i) A simple optical device consisting of a closed box with a small aperture on one side that forms a real, inverted image on a translucent screen without using any lenses.	A. Compound Microscope
(ii) An instrument that utilizes two parallel mirrors or prisms set at an angle to allow an observer inside a submarine to view objects above the water surface.	B. Simple Periscope
(iii) An optical instrument featuring a single converging lens used to produce an enlarged, upright, and virtual image of very small text or objects.	C. Astronomical Telescope
(iv) A dual-lens system designed to observe highly magnified images of minute specimens such as biological cells in a school laboratory.	D. Pinhole Camera

LIST A	LIST B
(v) An instrument optimized with specific lens configurations to view distinct details of highly distant celestial bodies like stars and planets.	E. Magnifying Glass
	F. Slide Projector
	G. Human Eye
	H. Camera Lens

ANSWER

i	ii	iii	iv	v

SECTION B (70 Marks)

Answer **all** questions in this section.

3. (a) For many centuries, scholars observed that heavy objects fell toward the Earth, but it was not until a structured framework was established that the underlying causes were organized into formal definitions. In science, ideas progress from initial thoughts to established rules.

(i) Distinguish clearly between a scientific theory and a scientific principle.

(ii) Explain how an initial hypothesis can eventually be accepted as a universally recognized scientific principle or law.

(c) A local farmer notices that the metal parts of tools stored in a damp shed develop a reddish-brown layer much faster than identical tools stored inside a dry farmhouse kitchen. The farmer decides to systematically investigate this observation to find a definitive cause.

(i) Identify the first step of the scientific method that the farmer automatically performed when noticing the difference between the tools.

(ii) Formulate a suitable, testable hypothesis based on the farmer's observation.

(iii) State the independent variable and the dependent variable if an experiment were to be set up to test this occurrence in a school laboratory.

4. A heavy wooden crate sits undisturbed on the flat floor of a school storage room. A student approaches the crate and attempts to push it across the floor by applying a horizontal push. Initially, despite the student pushing quite hard, the crate remains completely stationary. When a second student joins and they both push together with a much greater combined effort, the crate suddenly begins to slide forward smoothly across the room.

(i) Identify the two main types of forces acting on the crate along the vertical plane while it is sitting completely at rest before anyone touches it.

(ii) Explain why the crate did not move at all when the first student applied an initial horizontal pushing force.

(iii) State the term used to describe the state of the forces acting on the crate when it remains completely stationary despite being pushed.

(iv) Describe what happens to the net force acting on the crate when the second student joins, causing the crate to transition from rest into a state of uniform motion.

(v) Classify the forces involved in this entire scenario into either **contact forces** or **non-contact forces**.

5. (a) A large block of dry wood is placed in a large tank of water. It floats with a significant portion of its body remaining above the water level.

(i) State the relationship between the weight of the floating wood and the upthrust acting on it.

(ii) How does the weight of the water displaced by the wood compare to the total weight of the wood?

(iii) What would happen if the same block of wood were placed in a liquid that is much denser than water?

(b) A mechanic stretches a steel spring by pulling both ends. When the pulling force is released, the spring returns completely to its original length.

(i) Name the law that governs this behavior within the material's limits.

(ii) Describe what happens to the spring if it is stretched far beyond its elastic limit.

6. (a) A water supply company constructs a large distribution dam. The walls at the very bottom of the dam are built to be much thicker and stronger than the walls at the top.

(i) Explain why the design requires thicker walls at the bottom.

(ii) State how the density of the liquid affects the internal pressure at any given depth.

(iii) Why do deep-sea divers need to wear specially reinforced diving suits?

(b) A car moves along a straight, flat highway. The driver maintains a steady reading on the speedometer for several kilometers.

(i) Describe the state of acceleration of the car during this period.

(ii) Distinguish clearly between distance and displacement as applied to this motion.

7. (a) A weightlifter holds a heavy barbell stationary above their head for two full minutes. Despite sweating and feeling exhausted, the physics teacher states that the weightlifter did no work on the barbell during those two minutes.

(i) Justify the physics teacher's statement based on the scientific definition of work.

(ii) Explain the difference between the terms "energy" and "power."

(iii) Name the form of energy stored in the barbell while it is held high in the air.

(b) A solar panel is installed on the roof of a house to power a small water pump.

(i) State the main energy transformation that takes place in the solar panel.

(ii) Identify one factor that can reduce the operational efficiency of this energy system.

8.(a) A technician sets up a simple domestic circuit where three bulbs are connected such that if one bulb burns out or is removed, the remaining two bulbs instantly go dark.

(i) Identify the type of circuit arrangement used by the technician.

(ii) Explain the main disadvantage of using this specific arrangement in residential houses.

(iii) Mention the instrument used to detect the presence of an electric current in a circuit.

(b) An iron nail is rubbed repeatedly in one direction using the north pole of a strong permanent bar magnet.

(i) Name the method of magnetization described in this scenario.

(ii) Describe how you could test if the iron nail has successfully acquired magnetic properties.

9. (a) A person stands in front of a large, vertical plane mirror in a dressing room to inspect their outfit.

(i) State two distinct characteristics of the image formed by this plane mirror.

(ii) Why does the writing on the person's t-shirt appear reversed when viewed in the mirror reflection?

(iii) Name the phenomenon where light bounces off a rough, unpolished wall surface in many different directions.

(b) A student views a bright red dress under pure blue light illumination inside a dark theatre room.

(i) State the apparent color of the dress as seen by the student under this blue light.

(ii) Give a brief reason for your answer in (i).

SECTION C (15 Marks)

Answer **all** questions in this section.

10. A petroleum delivery truck is observed moving down a highway with a heavy metal chain attached to its chassis dragging continuously along the asphalt road surface.

(a) Analyze the physical cause of charge accumulation on the body of a moving fuel tanker.

(b) Explain the potential danger associated with leaving these accumulated static charges unchecked during transit or fuel offloading.

(c) Evaluate the role of the dragging metal chain in mitigating the dangers described in (b).

(d) Recommend two standard safety practices that fuel station attendants must observe when handling highly volatile fuels to prevent static electricity hazards.
